M 55/75

Enrolment Number										

Total No. of printed pages = 03

Monsoon, 2024

B. Tech 3rd Semester Examination

Mathematics To Civil Engineering

Course Code: BMA23211T

Full Marks – 50 **Time – 2 hours**

The figure in the margin indicates full marks for the questions.

Answer the following questions

1. Choose the correct options: 1X10=10

a) The period of the function $f(x) = sinx$ is
 (i) π
 (ii) 2π
 (iii) 3π
 (iv) 4π

b) The function $f(x) = cosx, 0 < x < \pi$ is
 (i) an odd function
 (ii) an even function
 (iii) neither an odd nor an even function
 (iv) both and odd and even function

c) The Fourier series of even function contains only
 (i) sine terms
 (ii) cosine terns
 (iii) both sine and cosine terms
 (iv) none of these

d) $L\{cos4t\} =$
 (i) $\frac{4}{s^2+16}$
 (ii) $\frac{s}{s^2+16}$
 (iii) $\frac{4}{s^2-16}$
 (iv) $\frac{4}{s^2-16}$

e) If $L\{f(t)\} = \bar{f}(s)$, then $L\{f'(t)\} =$
 (i) $\bar{f}(s) + f(0)$
 (ii) $\bar{f}(s) - sf(0)$
 (iii) $s\bar{f}(s) + f(0)$

(iv) $s\bar{f}(s) - f(0)$

f) If $L\{f(t)\} = \bar{f}(s)$, then $L\{f(at)\} =$

(i) $a\bar{f}(s)$

(ii) $\frac{1}{a}\bar{f}(s)$

(iii) $\frac{1}{a}\bar{f}\left(\frac{s}{a}\right)$

(iv) $a\bar{f}\left(\frac{s}{a}\right)$

g) If $L^{-1}\{\bar{f}(s)\} = f(t), L^{-1}\{\bar{g}(s)\} = gf(t)$, then $L^{-1}\{\bar{f}(s)\bar{g}(s)\} =$

(i) $\int_0^t f(u)g(u)du$

(ii) $\int_0^t f(u)g(t-u)du$

(iii) $\int_0^\infty f(u)g(u)du$

(iv) $\int_0^\infty f(u)g(t-u)du$

h) The degree of the partial differential equation $\left(\frac{\partial z}{\partial x}\right)^4 + \frac{\partial^3 z}{\partial y^3} = 2\frac{\partial z}{\partial x}$ is

(i) 0

(ii) 1

(iii) 2

(iv) 4

i) The one-dimensional Heat equation is given by

(i) $\frac{\partial u}{\partial x} = c^2\frac{\partial^2 u}{\partial t^2}$

(ii) $\frac{\partial u}{\partial t} = c^2\frac{\partial^2 u}{\partial x^2}$

(iii) $\frac{\partial^2 u}{\partial x^2} = c^2\frac{\partial^2 u}{\partial t^2}$

(iv) $\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}$

j) The solution of the one-dimensional wave equation takes the form of

(i) a logarithmic function

(ii) an algebraic function

(iii) a periodic function

(iv) none of these

2. a) Find a Fourier Series of the function $f(x) = x^2, -\pi \le x \le \pi$. Also deduce that

$\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots.. = \frac{\pi^2}{12}$ 8+2

OR

b) (i) Find a Fourier series representation of $f(x) = x, 0 \le x \le 2\pi$ 5

(ii) Find a Fourier cosine series $f(x) = e^x$ over $(0, 1)$. 5

3. a) Evaluate: (i) $L\{sin4tcos5t\}$ 4

(ii) $L\left\{\frac{cosat-cosbt}{t}\right\}$ 6

OR

b) Evaluate: (i) $L\{e^{-3t}(2cos5t - 3sin5t)\}$ 5

(ii) $L\{t^2 sint\}$ 5

4. a) Evaluate: (i) $L^{-1}\left\{\frac{6s-4}{s^2-4s+20}\right\}$ 3

(ii) $L^{-1}\left\{\frac{s}{(s^2+4)^2}\right\}$ using Convolution Theorem 7

OR

b) Solve the differential equation $Y'' - 3Y' + 2Y = 4e^{2t}$, given that $Y(0) = -3, Y'(0) = 5$ 10

5. a) A tightly stretched string of length l is initially in a position given by $y = asin^3 \frac{\pi x}{l}$. If it is released from rest from this position, find the displacement $y(x, t)$. 10

OR

b) Solve the one-dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ subject to boundary conditions $u(0, x) = 0 = u(l, t)$ and initial condition $u(x, 0) = u_0$

***